

Unnamed_Unit

Unit 3 - Creating Your First Program

PDF Documents

PDF Document 1: 1738219585298-Chapter 1 - Unit 3 - Creating Your First Program.pdf

This PDF document is included in the unit materials.



UNIT 3 CREATING YOUR FIRST PROGRAM

Now that you know how to give instructions and how computers follow commands, it's time to put everything together and create your very first program! In this unit, you'll learn how to combine instructions to make a character or robot complete a simple task.





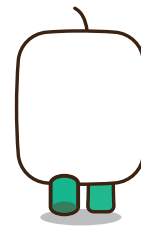
WHAT IS A PROGRAM?

A program is a set of instructions that a computer follows to do something. Just like when you follow steps to make a sandwich or build a toy, a computer follows the steps we give it. These steps are called commands, and when we put them in the right order, the computer can complete the task.

For example, if we want a robot to move through a maze, we'll give it commands like:



MOVE FORWARD



MOVE BACKWARD

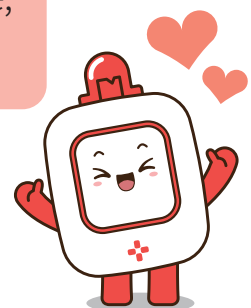


TURN LEFT



TURN RIGHT

By putting these commands together in a sequence, we can help the robot find its way out of the maze!



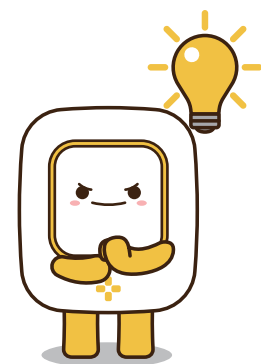
STEP-BY-STEP PROGRAMMING



When you write a program, it's important to put your commands in the right order. This is called sequencing. Just like when you get ready for school—first you put on your clothes, then your shoes, then grab your backpack—you need to put your commands in order so the computer knows what to do.

Here's an example of how sequencing works in programming:

- 1. Move forward** three steps.
- 2. Turn left.**
- 3. Move forward** two more steps.
- 4. Jump** over an obstacle.

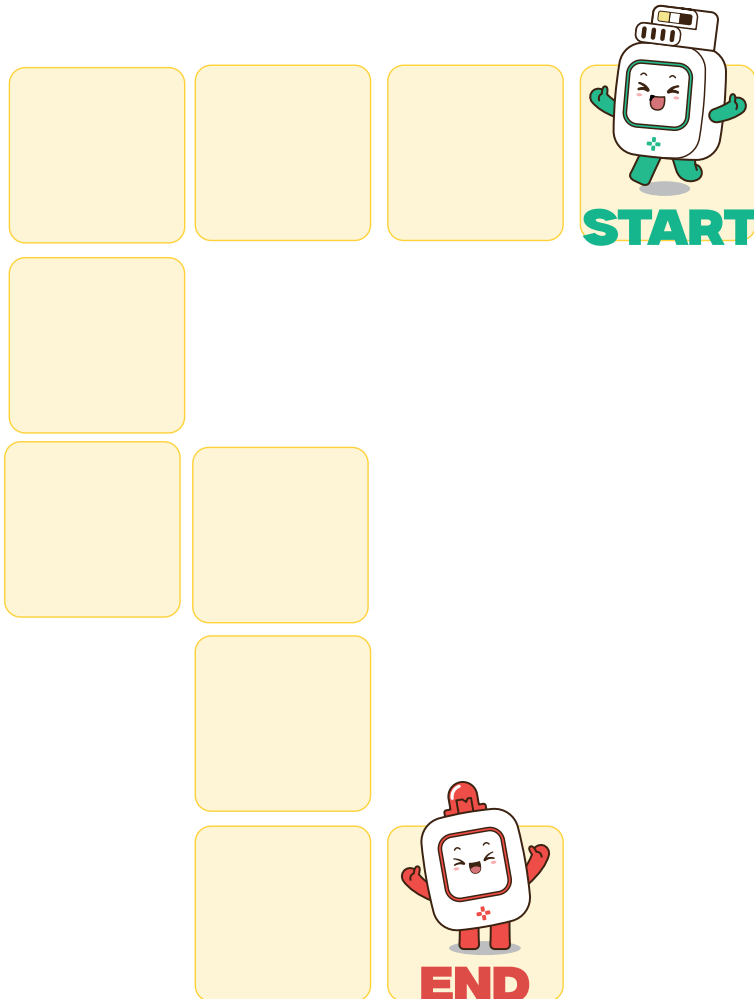




Let's practice



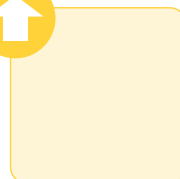
Let's figure out how many times Butt needs to move up, down, left, or right to meet Elly!



Let's practice



Let's figure out how many times Butt needs to move up, down, left, or right to meet Elly!





● Multiple Choice Questions

1 What is programming??

- Ⓐ A type of game
- Ⓑ Giving instructions to a computer
- Ⓒ Writing stories
- Ⓓ Writing stories

CORRECT ANSWER

Ⓑ

2 Which of these objects is the same color as the second color of the rainbow?

- Ⓐ Step-by-step in the right sequence
- Ⓑ Forward
- Ⓒ Jump
- Ⓓ Radom order

CORRECT ANSWER

Ⓐ

3 What should you do if your program doesn't work correctly?

- Ⓐ Plays music
- Ⓑ Fix the mistakes and try again
- Ⓒ Give up
- Ⓓ Leave it as it is

CORRECT ANSWER

Ⓑ

● Review Questions

- 1 Why is it important to give instructions in the right order when programming a computer?

- 2 What are some commands you might use to make a robot or character move in a program?